

Task 1: Design and Configure AWS Networking Infrastructure

Objective

Design a secure and scalable networking infrastructure using Amazon VPC.

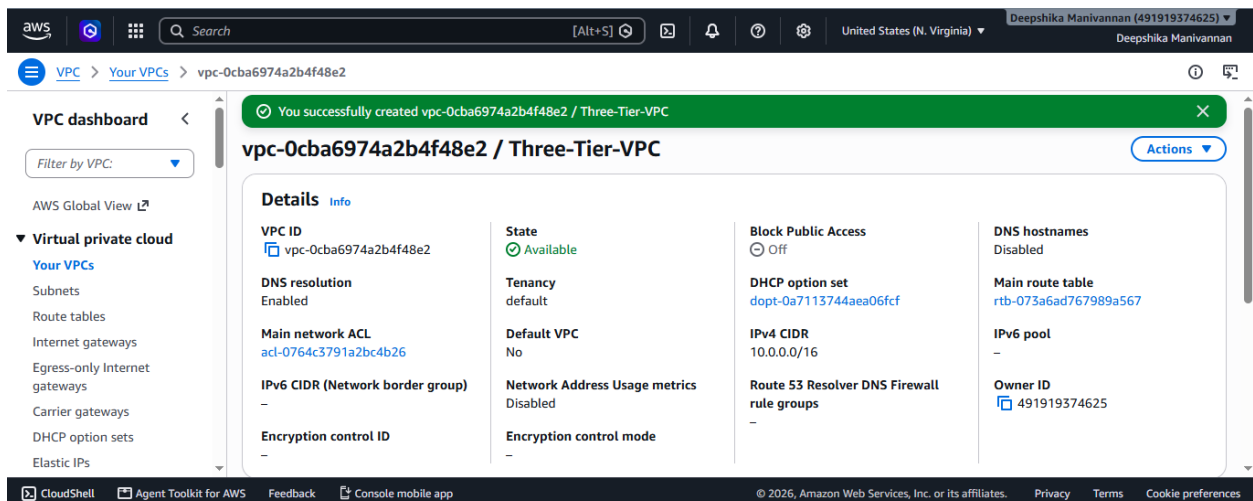
Virtual Private Cloud (VPC) : A VPC is a private network in AWS where all resources are deployed.

Why? To isolate our application infrastructure from other AWS customers and create our own networking environment.

Step 1: Create VPC

Configuration:

A custom VPC was created with CIDR block 10.0.0.0/16 to host the three-tier architecture.



Step 2: Create Public Subnets

Subnets: Subnets divide the VPC into smaller network segments. We created 6 subnets across 2 Availability Zones.

Why?

- Public Subnets host web servers.
- Private App Subnets host Flask application servers.
- Private DB Subnets host Amazon RDS.

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VPC > Subnets > subnet-0e58b03e057c61165

VPC dashboard < subnet-0e58b03e057c61165 / Public-Subnet-A

Filter by VPC: AWS Global View

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs

Details	
Subnet ID subnet-0e58b03e057c61165	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-0e58b03e057c61165
IPv4 CIDR 10.0.1.0/24	Available IPv4 addresses 251
Availability Zone use1-az1 (us-east-1a)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPv6 CIDR reservations	
State Available	IPv6 CIDR -
Block Public Access Off	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
IPv6 CIDR association ID -	Auto-assign public IPv4 address No
Route table -	Outpost ID -
Auto-assign IPv6 address No	Hostname type
IPv4 CIDR reservations -	
Resource name DNS A record Disabled	

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VPC > Subnets > subnet-00173f4a8df177c94

VPC dashboard < subnet-00173f4a8df177c94 / Public-Subnet-B

Filter by VPC: AWS Global View

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs

Details	
Subnet ID subnet-00173f4a8df177c94	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-00173f4a8df177c94
IPv4 CIDR 10.0.2.0/24	Available IPv4 addresses 251
Availability Zone use1-az2 (us-east-1b)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPv6 CIDR reservations	
State Available	IPv6 CIDR -
Block Public Access Off	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
IPv6 CIDR association ID -	Auto-assign public IPv4 address No
Route table -	Outpost ID -
Auto-assign IPv6 address No	Hostname type
IPv4 CIDR reservations -	
Resource name DNS A record Disabled	

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Step 3: Create Application Subnets

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VPC > Subnets > subnet-0e14474cd1d198328

VPC dashboard < subnet-0e14474cd1d198328 / App-Subnet-A

Filter by VPC: AWS Global View

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs

Details	
Subnet ID subnet-0e14474cd1d198328	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-0e14474cd1d198328
IPv4 CIDR 10.0.11.0/24	Available IPv4 addresses 251
Availability Zone use1-az1 (us-east-1a)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPv6 CIDR reservations	
State Available	IPv6 CIDR -
Block Public Access Off	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
IPv6 CIDR association ID -	Auto-assign public IPv4 address No
Route table -	Outpost ID -
Auto-assign IPv6 address No	Hostname type
IPv4 CIDR reservations -	
Resource name DNS A record Disabled	

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The screenshot shows the AWS Management Console interface for a VPC dashboard. The left sidebar contains navigation links for VPCs, Subnets, Route tables, Internet gateways, Egress-only Internet gateways, Carrier gateways, DHCP option sets, and Elastic IPs. The main content area displays the details for subnet-0a3906c0006987fdf / App-Subnet-B. The details are organized into a grid with the following information:

Details	
Subnet ID subnet-0a3906c0006987fdf	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-0a3906c0006987fdf
IPV4 CIDR 10.0.12.0/24	Available IPv4 addresses 251
Availability Zone use1-az2 (us-east-1b)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPV6 CIDR reservations -	State Available
	IPV6 CIDR -
	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
	Auto-assign public IPv4 address No
	Outpost ID -
	Hostname type -
Block Public Access Off	IPv6 CIDR association ID -
	Route table -
	Auto-assign IPv6 address No
	IPv4 CIDR reservations -
	Resource name DNS A record Disabled

Step 4: Create Database Subnets

The first screenshot shows the details for subnet-0a2e2a83eb32022ca / DB-Subnet-A. The details are organized into a grid with the following information:

Details	
Subnet ID subnet-0a2e2a83eb32022ca	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-0a2e2a83eb32022ca
IPV4 CIDR 10.0.21.0/24	Available IPv4 addresses 251
Availability Zone use1-az1 (us-east-1a)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPV6 CIDR reservations -	State Available
	IPV6 CIDR -
	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
	Auto-assign public IPv4 address No
	Outpost ID -
	Hostname type -
Block Public Access Off	IPv6 CIDR association ID -
	Route table -
	Auto-assign IPv6 address No
	IPv4 CIDR reservations -
	Resource name DNS A record Disabled

The second screenshot shows the details for subnet-0839791b02d22bcdf / DB-Subnet-B. The details are organized into a grid with the following information:

Details	
Subnet ID subnet-0839791b02d22bcdf	Subnet ARN arn:aws:ec2:us-east-1:49191937:4625:subnet/subnet-0839791b02d22bcdf
IPV4 CIDR 10.0.22.0/24	Available IPv4 addresses 251
Availability Zone use1-az2 (us-east-1b)	Network border group us-east-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPV6 CIDR reservations -	State Available
	IPV6 CIDR -
	VPC vpc-0c6a6974a2b4f48e2 Three-Tier-VPC
	Auto-assign public IPv4 address No
	Outpost ID -
	Hostname type -
Block Public Access Off	IPv6 CIDR association ID -
	Route table -
	Auto-assign IPv6 address No
	IPv4 CIDR reservations -
	Resource name DNS A record Disabled

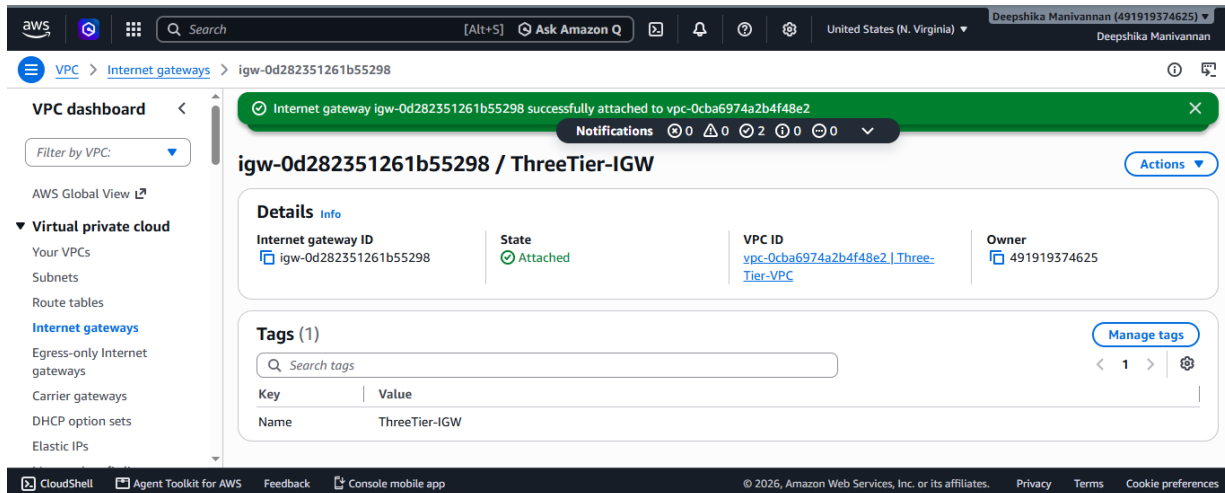
Step 5: Create Internet Gateway

Internet Gateway (IGW) : A gateway that provides internet access to public resources.

Why?

Without an Internet Gateway:

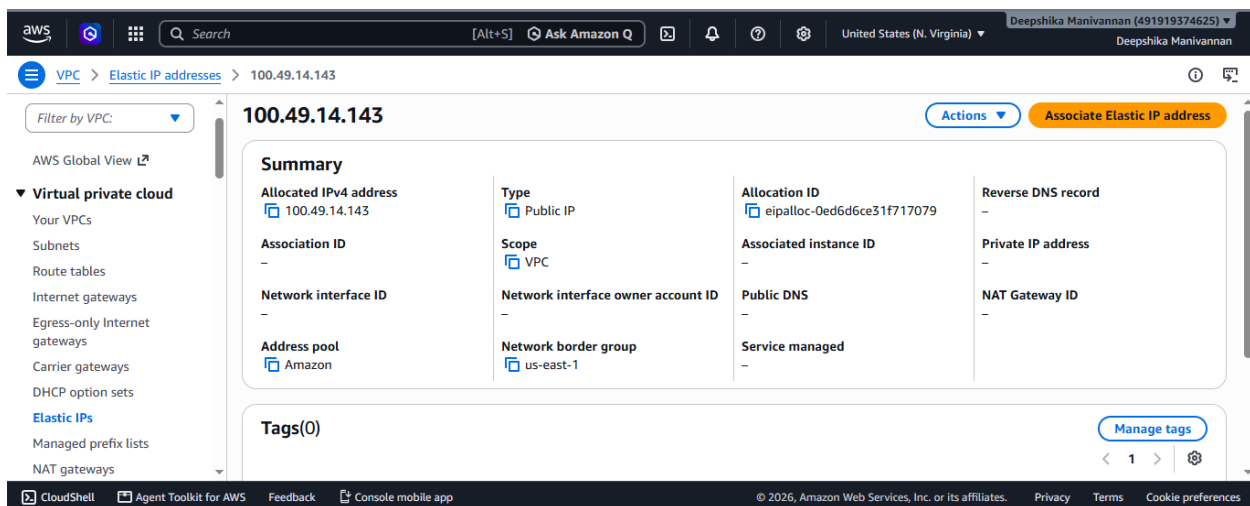
- ✗ Users cannot access the website.
- ✗ EC2 instances cannot communicate with the internet.



Step 6: Allocate Elastic IP

Elastic IP (EIP) : A static public IP address provided by AWS.

Why? Used by the NAT Gateway to communicate with the internet.



Step 7: Create NAT Gateway

NAT Gateway: A managed AWS service that allows instances in private subnets to access the internet.

Why? Application servers may need:

1. Package installation
2. Software updates

NAT gateway nat-028204b15492ed579 / nat_three_tier

Details

NAT gateway ID nat-028204b15492ed579	Connectivity type Public	State Pending	State message Info -
NAT gateway ARN arn:aws:ec2:us-east-1:491919374625:natgateway/nat-028204b15492ed579	Primary public IPv4 address -	Primary private IPv4 address -	Primary network interface ID -
VPC vpc-0cba6974a2b4f48e2 / Three-Tier-VPC	Subnet subnet-0e58b03e057c61165 / Public-Subnet-A	Created Sunday, 20 September 2026 at 15:22:01 GMT+5:30	Deleted -

Secondary IPv4 addresses | **Monitoring** | **Tags**

Step 8: Create Route Tables

Route Tables: Route tables determine where network traffic is sent.

Public Route Table – Allows public subnet resources to access the internet, 0.0.0.0/0 → Internet Gateway. Connect to public subnet

Application Route Table - App servers remain private but can access the internet via NAT Gateway, 0.0.0.0/0 → NAT Gateway. Connect to private Appsubnet

Database Route Table - Keeps database resources private.

Public RT

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Route tables

Details

Route table ID rtb-056f5807ca4b6ae1e	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-0cba6974a2b4f48e2 / Three-Tier-VPC	Owner ID 491919374625		

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0d282351261b55298	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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VPC

Route tables

rtb-056f5807ca4b6ae1e

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Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

You have successfully updated subnet associations for rtb-056f5807ca4b6ae1e / Public_rt.

rtb-056f5807ca4b6ae1e

No

2 subnets

VPC

vpc-0cba6974a2b4f48e2 | Three-Tier-VPC

Owner ID

491919374625

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (2)

Find subnet association

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public-Subnet-A	subnet-0e58b03e057c61165	10.0.1.0/24	-
Public-Subnet-B	subnet-00173f4a8df177c94	10.0.2.0/24	-

Edit subnet associations

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VPC

Route tables

rtb-012804f77f7632f8d

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Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Details info

Route table ID

rtb-012804f77f7632f8d

Main

No

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Filter routes

Both

Edit routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-028204b15492ed579	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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VPC

Route tables

rtb-012804f77f7632f8d

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Route tables

Internet gateways

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Carrier gateways

DHCP option sets

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Peering connections

Route servers

Route table ID

rtb-012804f77f7632f8d

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Explicit subnet associations (2)

Find subnet association

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
App-Subnet-A	subnet-0e14474cd1d198328	10.0.11.0/24	-
App-Subnet-B	subnet-0a3906c0006987fdf	10.0.12.0/24	-

Edit subnet associations

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VPC > Route tables > rtb-032daf47e98139874

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Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Details Info

Route table ID [rtb-032daf47e98139874](#)

Main [No](#)

Explicit subnet associations [-](#)

Edge associations [-](#)

VPC [vpc-0cba6974a2b4f48e2 | Three-Tier-VPC](#)

Owner ID [491919374625](#)

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (1)

Destination	Target	Status	Propagated	Route Origin
10.0.0/16	local	Active	No	Create Route Table

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VPC > Route tables > rtb-032daf47e98139874

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Your VPCs
Subnets
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Egress-only Internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

You have successfully updated subnet associations for rtb-032daf47e98139874 / DB_rt.

VPC [vpc-0cba6974a2b4f48e2 | Three-Tier-VPC](#)

Owner ID [491919374625](#)

Subnet associations Routes Edge associations Route propagation Tags Related resources

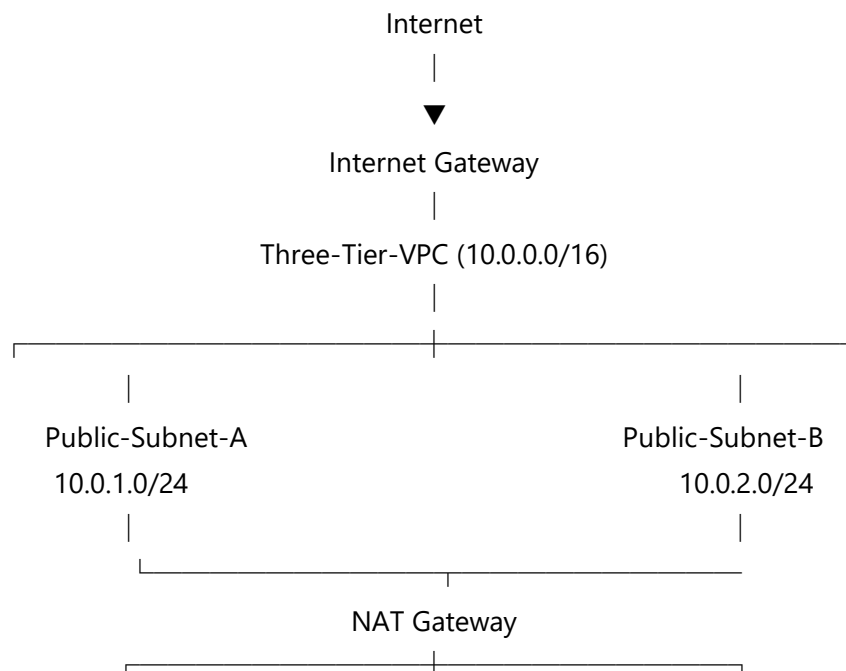
Explicit subnet associations (2)

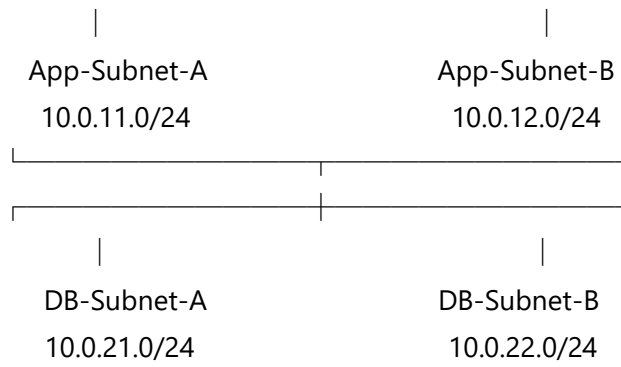
Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
DB-Subnet-A	subnet-0a2e2a83eb32022ca	10.0.21.0/24	-
DB-Subnet-B	subnet-0839791b02d22bcd	10.0.22.0/24	-

Subnets without explicit associations (0)

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Architecture:





Task 2 - Deploy the Web Tier (Presentation Layer)

Objective

Deploy web servers in public subnets and host the frontend application using Nginx.

Step 1: Launch Web Server 1 and verifying the Instance

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EC2 > Instances > i-0001c2765fd7231f2

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager

Instance summary for i-0001c2765fd7231f2 (web-server-1) Info

Updated 2 minutes ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0001c2765fd7231f2	Public IPv4 address 18.207.108.168 open address	Private IPv4 addresses 10.0.1.163
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-163.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-163.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recomm
Auto-assigned IP address 18.207.108.168 [Public IP]	VPC ID vpc-0cba6974a2b4f48e2 (Three-Tier-VPC) open	

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EC2 > Instances > i-0001c2765fd7231f2

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager

Instance summary for i-0001c2765fd7231f2 (web-server-1) Info

Updated 2 minutes ago

[Connect](#) [Instance state](#) [Actions](#)

Details Status and alarms Monitoring **Security** Networking Storage Tags

▼ Security details

IAM role -	Owner ID 491919374625	Launch time Tue Sep 22 2026 05:35:55 GMT+0530 (India Standard Time)
Security groups sg-0cc4c7e7c8a453493 (web_sg)		

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source
1				

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```
SSH session to ubuntu@18.207.108.168
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✓ (remote display is forwarded through SSH)
For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Tue Sep 22 01:33:39 UTC 2026

System load: 0.0 Temperature: -273.1 C
Usage of /: 30.7% of 6.61GB Processes: 127
Memory usage: 28% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 10.0.1.163

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Sep 22 00:52:04 2026 from 103.66.77.233
ubuntu@ip-10-0-1-163:~$
```

ip-10-0-1-163 0% 0.33 GB / 0.89 GB 0.02 Mb/s 0.00 Mb/s 87 min /: 31%

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Launch Web Server 2 and verifying the Instance

Instance summary for i-0001c2765fd7231f2 (web-server-1) Info

Updated less than a minute ago

Instance ID i-0001c2765fd7231f2	Public IPv4 address 18.207.108.168 open address	Private IPv4 addresses 10.0.1.163
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-1-163.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-1-163.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recomm
Auto-assigned IP address 18.207.108.168 [Public IP]	VPC ID vpc-0cba6974a2b4f48e2 (Three-Tier-VPC)	

Security details

IAM role -	Owner ID 491919374625	Launch time Tue Sep 22 2026 05:35:55 GMT+0530 (India Standard Time)
Security groups sg-0cc4c7e7c8a453493 (web_sg)		
Inbound rules Filter rules		

```
2. 18.207.108.168 x 4. 3.87.161.148
SSH session to ubuntu@3.87.161.148
• Direct SSH : ✓
• SSH compression : ✓
• SSH-broker : ✓
• X11-forwarding : ✓ (remote display is forwarded through SSH)
• For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
 * Documentation: https://docs.ubuntu.com
 * Management: https://landscape.canonical.com
 * Support: https://ubuntu.com/pro

System information as of Tue Sep 22 01:32:28 UTC 2026
System load: 0.01 Temperature: -273.1 C
Usage of /: 30.5% of 6.61GB Processes: 122
Memory usage: 20% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 10.0.2.93

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

/usr/bin/xaauth: file /home/ubuntu/.xaauthority does not exist
ubuntu@ip-10-0-2-93:~$
```

Step 2: Install Nginx in both EC2

Command used:

Sudo apt update

Sudo apt install nginx -y

Verified the installation using: nginx -v

```
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 181
Download size: 666 kB
Space needed: 1880 kB / 4656 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.11
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.11 [628
Fetched 666 kB in 0s (21.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack ../nginx-common_1.28.3-2ubuntu1.11_all.deb ...
Unpacking nginx-common (1.28.3-2ubuntu1.11) ...
Selecting previously unselected package nginx.
Preparing to unpack ../nginx_1.28.3-2ubuntu1.11_amd64v3.deb ...
Unpacking nginx (1.28.3-2ubuntu1.11) ...
Setting up nginx-common (1.28.3-2ubuntu1.11) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.11) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-163:~$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
ubuntu@ip-10-0-1-163:~$
```

Not secure 18.207.108.168

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 Begin browsing instantly: Chrome can now launch when Windows starts. Allow

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.11 [37.9 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.11 [628 kB]
Fetched 666 kB in 0s (25.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.28.3-2ubuntu1.11_all.deb ...
Unpacking nginx-common (1.28.3-2ubuntu1.11) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.28.3-2ubuntu1.11_amd64v3.deb ...
Unpacking nginx (1.28.3-2ubuntu1.11) ...
Setting up nginx-common (1.28.3-2ubuntu1.11) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.11) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-2-93:~$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
ubuntu@ip-10-0-2-93:~$
```

ip-10-0-2-93 0% 0.32 GB / 0.89 GB 0.01 Mb/s 0.00 Mb/s 102 min /: 35% /tmp: 0% /boot: 11% /boot/efi: 7%

Not secure 3.87.161.148

VL_Zen OneNote Deepshikamanivann... Devops - Kanban B... Module 1 - Linux j... Module wise practic... GitHub - iam-veera...

Begin browsing instantly: Chrome can now launch when Windows starts. [Allow](#)

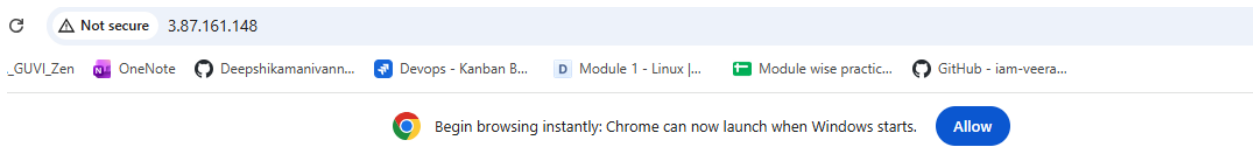
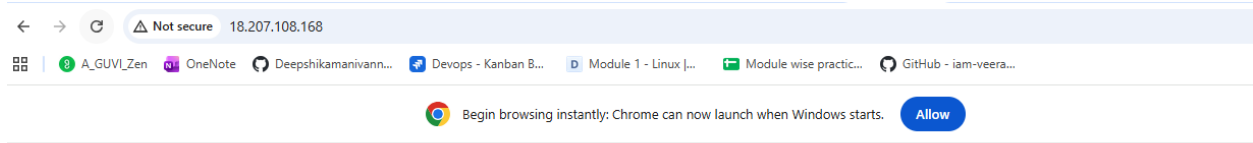
Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 3: Modify the ihtml index



Task 3 – Deploy the Application Tier

Objective

Deploy the backend Flask application on EC2 instances located in private subnets.

The Application Tier will:

Process requests coming from the Web Tier.

Connect to Amazon RDS.

Store and retrieve application data.

Architecture:

Web Tier (Nginx)

|



App Server 1 (Flask)

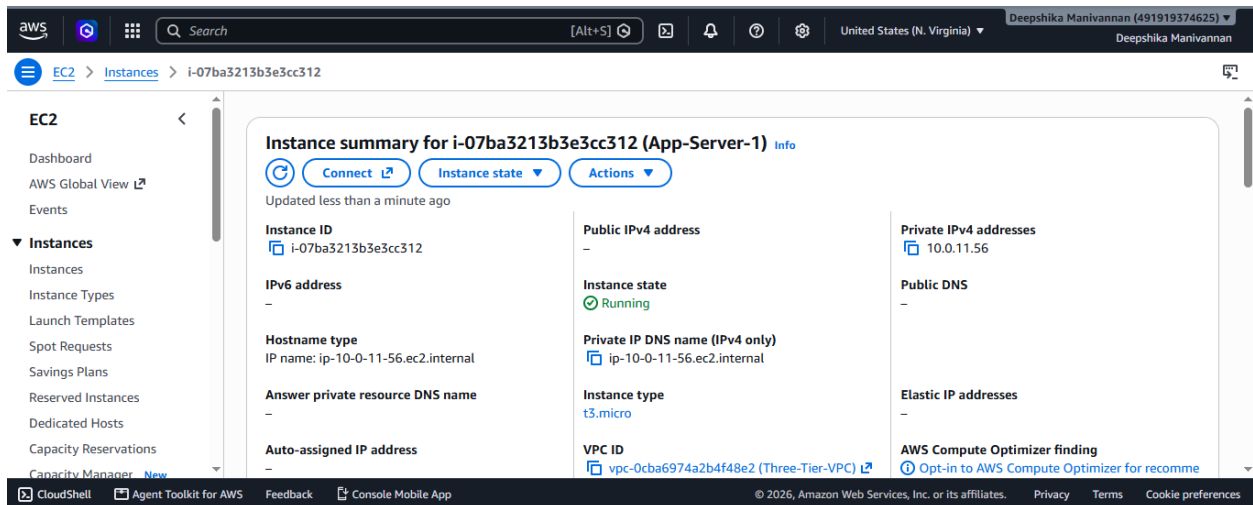
App Server 2 (Flask)

|

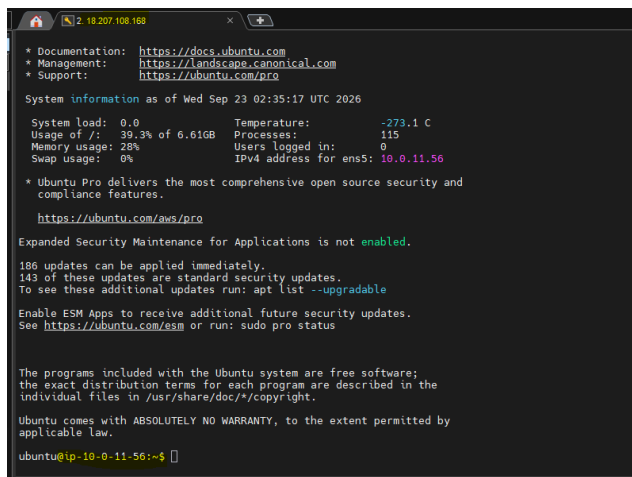


Amazon RDS

EC2-Private App server –1



Connect to App server via web server using scp



EC2-Private App server –2

aws [Search] [Alt+S] United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Instances > i-0c2c99d10a77b93fc

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager

Instance summary for i-0c2c99d10a77b93fc (App-Server-2) Info

Updated 3 minutes ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0c2c99d10a77b93fc	Public IPv4 address -	Private IPv4 addresses 10.0.12.178
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-12-178.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-12-178.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommen
Auto-assigned IP address -	VPC ID vpc-0c6a6974a2b4f48e2 (Three-Tier-VPC)	

Private IPv4 address copied

Connect to App server via web server using scp

```

* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Wed Sep 23 03:15:49 UTC 2026

System load: 0.0      Temperature: -273.1 C
Usage of /: 39.3% of 6.61GB   Processes: 115
Memory usage: 28%      Users logged in: 0
Swap usage: 0%          IPv4 address for ens5: 10.0.12.178

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

186 updates can be applied immediately.
142 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-12-178:~$

```

ip-10-0-2-93 0% 0.34 GB / 0.89 GB 0.01 Mb/s 0.00 Mb/s 27 hours /: 41% /tmp: 0% /boot: 18% /boot/

Step 2: Configure Security Group

The screenshot shows the AWS Management Console interface. The left sidebar displays the navigation menu with 'EC2' selected. The main content area shows the 'Security Groups' page for the group 'sg-0a984deeb1e5aba9b'. The 'Inbound rules' tab is active, showing two rules: 'SSH' (Type: SSH, Protocol: TCP, Port range: 22) and 'Custom TCP' (Type: Custom TCP, Protocol: TCP, Port range: 5000). The 'Outbound rules' tab is also visible, showing one rule. The 'Sharing' tab is active, showing the group is shared with the VPC 'vpc-0cba6974a2b4f48e2'. The 'Related resources' tab is also visible.

Run Flask Application

App-1

```
'cursorclass': pymysql.cursors.DictCursor
}

@app.route('/')
def health():
    return jsonify({'status': 'Flask is running'})

@app.route('/submit', methods=['POST'])
def submit():
    data = request.get_json()
    conn = pymysql.connect(**DB_CONFIG)
    with conn.cursor() as cur:
        cur.execute(
            'INSERT INTO feedback (name, email, feedback) VALUES (%s, %s, %s)',
            (data['name'], data['email'], data['feedback'])
        )
    conn.commit()
    conn.close()
    return jsonify({'message': 'Feedback submitted successfully!'})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

(venv) ubuntu@ip-10-0-11-56:~$ python3 app.py &
[1] 5595
(venv) ubuntu@ip-10-0-11-56:~$ * Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.0.11.56:5000
Press CTRL+C to quit
^C
(venv) ubuntu@ip-10-0-11-56:~$ curl localhost:5000
127.0.0.1 - - [23/Sep/2026 04:33:54] "GET / HTTP/1.1" 200 -
{"status": "Flask is running"}
(venv) ubuntu@ip-10-0-11-56:~$
```

App-2

```
2. 18.207.108.168 x 3. 3.87.161.148 x +
}
'cursorclass': pymysql.cursors.DictCursor
}

@app.route('/')
def health():
    return jsonify({'status': 'Flask is running'})

@app.route('/submit', methods=['POST'])
def submit():
    data = request.get_json()
    conn = pymysql.connect(**DB_CONFIG)
    with conn.cursor() as cur:
        cur.execute(
            'INSERT INTO feedback (name, email, feedback) VALUES (%s, %s, %s)',
            (data['name'], data['email'], data['feedback'])
        )
    conn.commit()
    conn.close()
    return jsonify({'message': 'Feedback submitted successfully!'})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
(venv) ubuntu@ip-10-0-12-178:~$ python3 app.py &
[1] 5559
(venv) ubuntu@ip-10-0-12-178:~$ * Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.0.12.178:5000
Press CTRL+C to quit
^C
(venv) ubuntu@ip-10-0-12-178:~$ curl localhost:5000
127.0.0.1 - - [23/Sep/2026 04:49:58] "GET / HTTP/1.1" 200 -
{"status": "Flask is running"}
(venv) ubuntu@ip-10-0-12-178:~$
```

ip-10-0-2-93 0% 0.35 GB / 0.89 GB 0.01 Mb/s 0.00 Mb/s 28 hours /: 41% /tmp: 0% /boot: 18% /boot/efi: 7%

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Task 4 Objective

Create an Amazon RDS MySQL database and connect it to the Flask application.

Architecture after Task 4:

Step 1: Create RDS Subnet Group

The screenshot shows the AWS Management Console interface for the 'three-tier-db-subnet-group' subnet group. The left sidebar contains navigation links for Aurora and RDS, Subnet groups, Parameter groups, Option groups, Custom engine versions, Zero-ETL integrations, Events, Event subscriptions, Recommendations, and Certificate update. The main content area displays the subnet group details, including its ARN, supported network types (IPv4), and a description. Below this, a table lists the subnets within the group.

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	DB-Subnet-A	subnet-0a2e2a83eb32022ca	10.0.21.0/24
us-east-1b	DB-Subnet-B	subnet-0839791b02d22bcd	10.0.22.0/24

Step 2: Create MySQL RDS

The screenshot shows the AWS Management Console for the 'three-tier-mysql' RDS instance. The left sidebar lists navigation options under 'Aurora and RDS', including 'Databases', 'Snapshots', 'Exports in Amazon S3', 'Automated backups', 'Reserved instances', 'Proxies', 'Subnet groups', 'Parameter groups', and 'Option groups'. The main content area displays the instance details:

- Summary**
 - DB identifier: three-tier-mysql
 - Status: Available
 - Role: Instance
 - Engine: MySQL Community
 - CPU: -
 - Class: db.t4g.micro
 - Current activity: -
 - Region & AZ: us-east-1b
- Recommendations**
- Connectivity & security** (selected tab)
- Monitoring**
- Logs & events**
- Configuration**
- Zero-ETL integrations**
- Maintenance**

At the bottom, there are links for 'Connect using' (Code snippets, CloudShell, Endpoints) and a footer with '© 2026, Amazon Web Services, Inc. or its affiliates.' and 'Privacy Terms Cookie preferences'.

Step 3: Create DB Security Group

The screenshot shows the AWS Management Console for the 'sg-05687388287125647' DB Security Group. The left sidebar lists navigation options under 'VPC', including 'gateways', 'Carrier gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', 'Peering connections', 'Route servers', 'Security', 'Network ACLs', 'Security groups', 'PrivateLink and Lattice', 'Getting started', 'Endpoints', and 'Endpoint services'. The main content area displays the security group details:

- Inbound security group rules successfully modified on security group (sg-05687388287125647 | DB-SG)**
- Details**
- Owner**: 491919374625
- Inbound rules count**: 1 Permission entry
- Outbound rules count**: 1 Permission entry
- Inbound rules** (selected tab)
- Outbound rules**
- Sharing**
- VPC associations**
- Related resources - new**
- Tags**
- Inbound rules (1)**
 - Rule ID: sg-0a984deeb1e5aba9...
 - IP version: -
 - Type: MYSQL/Aurora
 - Protocol: TCP
 - Port range: 3306
 - Source: sg-0a984deeb1e5aba9...

At the bottom, there are links for 'Manage tags' and 'Edit inbound rules', and a footer with '© 2026, Amazon Web Services, Inc. or its affiliates.' and 'Privacy Terms Cookie preferences'.

Step 4: Connect from App Server -1 to Db server

The screenshot shows a terminal window with the following commands and output:

```
ubuntu@ip-10-0-11-56:~$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 161.5k 100 161.5k 0 0 6.60M 0 0 0
ubuntu@ip-10-0-11-56:~$ mysql -h three-tier-mysql.car20igsm47t.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 36
Server version: 8.4.9 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Step 5: Create Database

```
3. 18.207.108.168 x 4. 18.207.108.168 +
-> created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
ERROR 1046 (3D000): No database selected
mysql> USE feedbackdb;
Database changed
mysql> CREATE TABLE feedback (
-> id INT AUTO_INCREMENT PRIMARY KEY,
-> name VARCHAR(100),
-> email VARCHAR(100),
-> feedback TEXT,
-> created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_feedbackdb |
+-----+
| feedback              |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE feedback;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int  | NO   | PRI | NULL    | auto_increment |
| name  | varchar(100) | YES |     | NULL    |                |
| email | varchar(100) | YES |     | NULL    |                |
| feedback | text | YES |     | NULL    |                |
| created | timestamp | YES |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.04 sec)

mysql> exit
Bye
```

```
3. 18.207.108.168 x 4. 18.207.108.168 +
{"feedback": "Hello RDS!"
},
{"message": "Feedback submitted successfully!"}
ubuntu@ip-10-0-11-56:~$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 161.5k 100 161.5k 0 0 8.63M 0 Total Spent Left Speed
0
ubuntu@ip-10-0-11-56:~$ mysql -h three-tier-mysql.car20igs47t.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY
--ssl-ca=/global-bundle.pem
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 56
Server version: 8.4.9 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE feedbackdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
+-----+-----+-----+-----+-----+
| id | name | email | feedback | created |
+-----+-----+-----+-----+-----+
| 1 | Test | t@t.com | Hello RDS! | 2026-09-24 04:21:04 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Step 4: Connect from App Server -2 to Db server

```
}
{"message": "Feedback submitted successfully!"}
ubuntu@ip-10-0-12-178:~$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total % Received % Xferd Average Speed Time Time Current
Dload Upload Total Spent Left Speed
100 161.5k 100 161.5k 0 0 5.71M 0 0 0 0
ubuntu@ip-10-0-12-178:~$ mysql -h three-tier-mysql.car2oigsm47t.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=/global-bundle.pem
Enter password:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 69
Server version: 8.4.9 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE feedbackdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM feedback;
+----+-----+-----+-----+-----+
| id | name | email | feedback | created |
+----+-----+-----+-----+-----+
| 1 | Test | t@t.com | Hello RDS! | 2026-09-24 04:21:04 |
| 2 | Test | t@t.com | Hello RDS! | 2026-09-24 04:43:14 |
+----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

Task 5 - Implement Network and Instance Security

Objective

Secure communication between tiers so that:

Web-sg

The screenshot shows the AWS Management Console interface. The left-hand navigation pane is expanded to 'Network & Security', with 'Security Groups' selected. The main content area displays the details for a security group named 'sg-0cc4c7e7c8a453493 - web_sg'. The 'Details' section shows the security group name, ID, description, VPC ID, owner, and rule counts. Below this, the 'Inbound rules' tab is active, showing a table with 5 rules. The table has columns for Name, Security group rule ID, IP version, Type, Protocol, and Port. The first rule is highlighted.

Name	Security group rule ID	IP version	Type	Protocol	Port

App-sg

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-0a984deeb1e5aba9b

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-0a984deeb1e5aba9b - App-sg

Actions

Details

Security group name App-sg	Security group ID sg-0a984deeb1e5aba9b	Description Creating sg for App	VPC ID vpc-0cba6974a2b4f48e2
Owner 491919374625	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port
------	------------------------	------------	------	----------	------

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In bound rules:

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-0a984deeb1e5aba9b

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-0a984deeb1e5aba9b - App-sg

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

Search

Type	Protocol	Port range	Source	Description
SSH	TCP	22	sg-0cc4c7e7c8a453493...	-
Custom TCP	TCP	5000	sg-0cc4c7e7c8a453493...	-

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aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-0cc4c7e7c8a453493

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-0cc4c7e7c8a453493

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (5) Manage tags Edit inbound rules

Search

Type	Protocol	Port range	Source	Description
SSH	TCP	22	144.16.179.199/32	-
HTTP	TCP	80	sg-0197254d83c87c05...	-
SSH	TCP	22	sg-0cc4c7e7c8a453493...	-
HTTP	TCP	80	0.0.0.0/0	-
HTTPS	TCP	443	0.0.0.0/0	-

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DB-sg

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-05687388287125647

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-05687388287125647 - DB-SG

Actions

Details

Security group name DB-SG	Security group ID sg-05687388287125647	Description Created by RDS management console	VPC ID vpc-0cba6974a2b4f48e2
Owner 491919374625	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1) Manage tags Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port

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aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-05687388287125647

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-05687388287125647 - DB-SG

Details

Security group name DB-SG	Security group ID sg-05687388287125647	Description Created by RDS management console	VPC ID vpc-0cba6974a2b4f48e2
Owner 491919374625	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1) Manage tags Edit inbound rules

Version	Type	Protocol	Port range	Source	Description
	MySQL/Aurora	TCP	3306	sg-0a984deeb1e5aba9...	-

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Public ALB

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Deepshika Manivannan (491919374625) Deepshika Manivannan

EC2 > Security Groups > sg-0197254d83c87c05e

Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

▼ Auto Scaling

- Auto Scaling Groups

sg-0197254d83c87c05e - Public_ALB-SG

Actions

Details

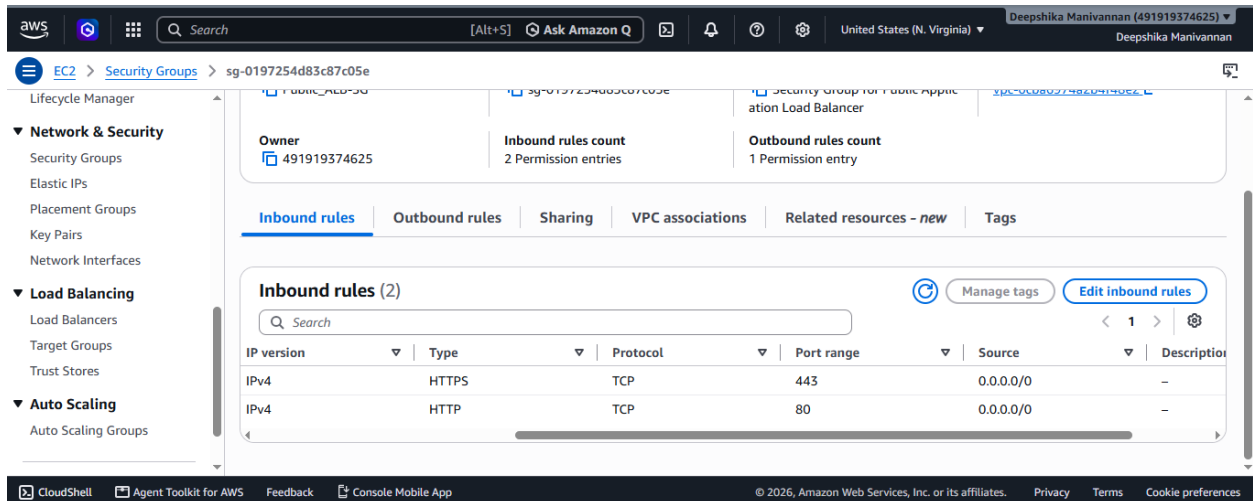
Security group name Public_ALB-SG	Security group ID sg-0197254d83c87c05e	Description Security Group for Public Application Load Balancer	VPC ID vpc-0cba6974a2b4f48e2
Owner 491919374625	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

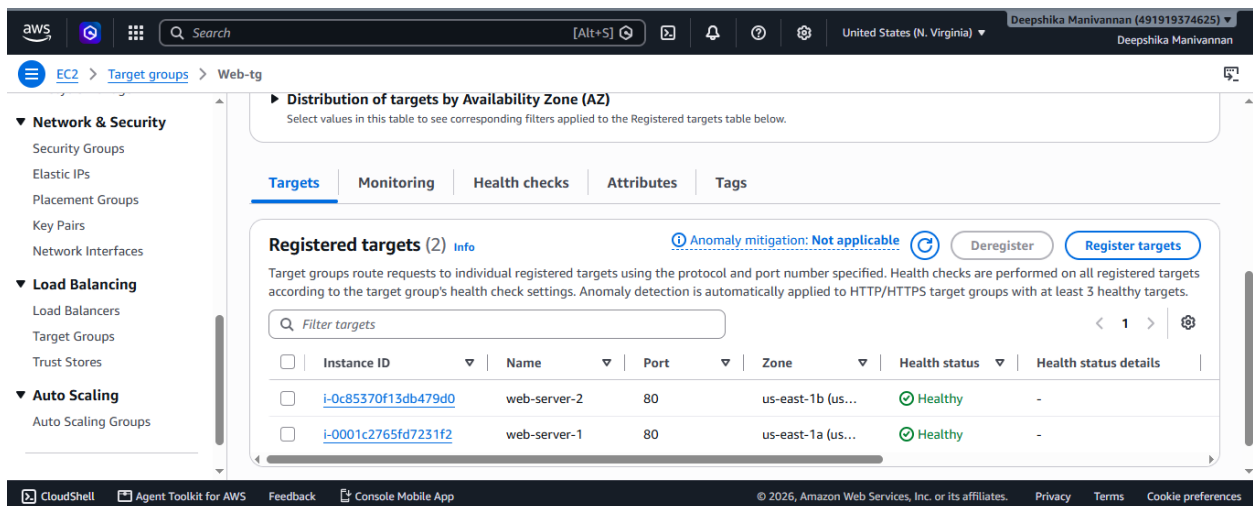
Name	Security group rule ID	IP version	Type	Protocol	Port

CloudShell Agent Toolkit for AWS Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

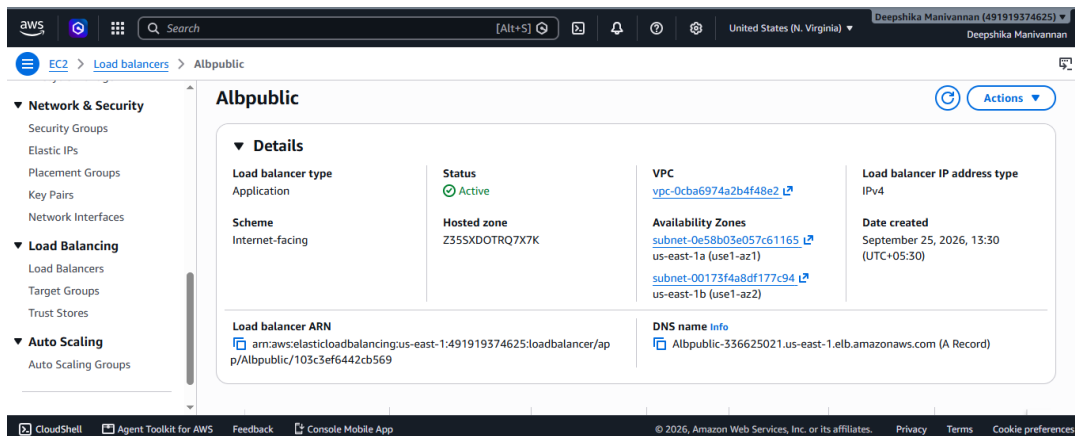


Task 06 – Internet-Facing Application Load Balancer

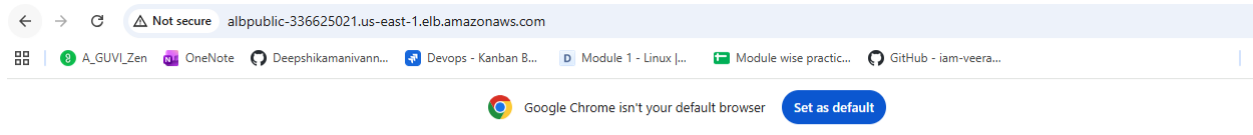
Step1 : Target Group Creation



Step2: creating PublicALB



Step 3: Open in Web browser using Public ALB



Feedback Form-2

Your Name

Your Email

Your Feedback

Submit

Task 7 – Configure Internal Application Load Balancer

Step 1: Create Target Group

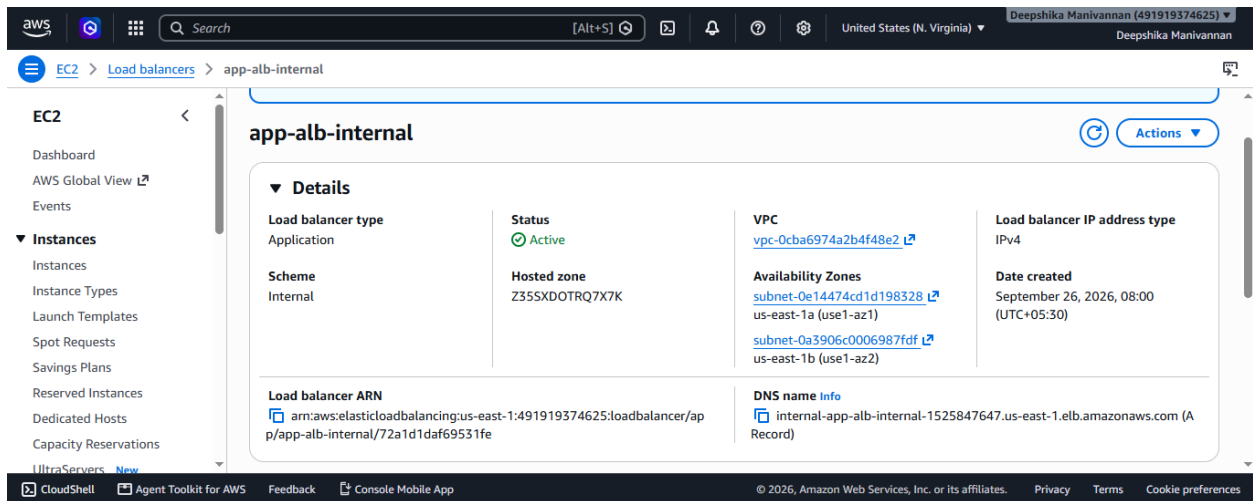
Registered targets (2) [Info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details
☐	i-0c2c99d10a77b93fc	App-Server-2	5000	us-east-1b (us...)	Healthy	-
☐	i-07ba3213b3e3cc312	App-Server-1	5000	us-east-1a (us...)	Healthy	-

Step 2: Internal ALB



Step3: connectivity check

```
ubuntu@ip-10-0-1-163:~$ curl http://10.0.11.56:5000
{"status": "Flask is running"}
ubuntu@ip-10-0-1-163:~$ curl http://internal-app-alb-internal-1525847647.us-east-1.elb.amazonaws.com
{"status": "Flask is running"}
ubuntu@ip-10-0-1-163:~$

ubuntu@ip-10-0-2-93:~$ curl http://10.0.12.178:5000
{"status": "Flask is running"}
ubuntu@ip-10-0-2-93:~$ curl http://internal-app-alb-internal-1525847647.us-east-1.elb.amazonaws.com
{"status": "Flask is running"}
ubuntu@ip-10-0-2-93:~$
```

The terminal output shows successful connectivity checks from two different EC2 instances (ip-10-0-1-163 and ip-10-0-2-93) to the internal load balancer. Both instances return the response {"status": "Flask is running"}.

Task 8 - Configure Nginx Reverse Proxy for API Routing

```
server {
    listen 80;

    server_name _;

    root /usr/share/nginx/html;

    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location /api/ {
        proxy_pass http://internal-app-alb-internal-1525847647.us-east-1.elb.amazonaws.com/; # Replace your Internal Loadbalancer DN
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}

ubuntu@ip-10-0-1-163:/etc/nginx$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-163:/etc/nginx$ sudo systemctl restart nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload u
nits.
ubuntu@ip-10-0-1-163:/etc/nginx$ sudo systemctl status nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-09-26 08:44:00 UTC; 8s ago
   Invocation: 48f561f3df184c59a00de7a9ea3e4ff5
```

```
worker_connections 1024;

http {
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    server {
        listen 80;

        server_name _;

        root /usr/share/nginx/html;

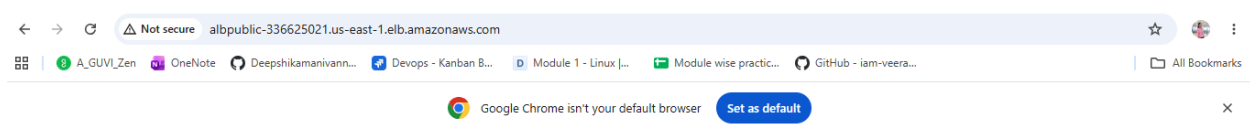
        index index.html;

        location / {
            try_files $uri $uri/ =404;
        }

        location /api/ {
            proxy_pass http://internal-app-alb-internal-1525847647.us-east-1.elb.amazonaws.com/; # Replace your Internal Loadbalancer DN
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
        }
    }
}

ubuntu@ip-10-0-2-93:/etc/nginx$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-2-93:/etc/nginx$
```

Task 9 - Integrate the Flask Application with Amazon RDS



Feedback Form-1

Feedback submitted successfully!

```
mysql> SELECT * FROM feedback;
```

id	name	email	feedback	created
1	Test	t@t.com	Hello RDS!	2026-09-24 04:21:04
2	Test	t@t.com	Hello RDS!	2026-09-24 04:43:14
3	deepshika	deepshika@gmail.com	testing	2026-09-26 09:02:30
4	deepshika	deepshika@gmail.com	testing	2026-09-26 09:02:58

```
4 rows in set (0.00 sec)
```